7inch DSI Display

User Manual

1. Overview

- 800x480 resolution capacitive touch display screen
- Raspberry Pi MIPI DSI interface for direct output, plug and play, without the need for driver installation
- Support Raspberry Pi OS, the official system of Raspberry Pi, with 2-finger zoom for web pages
- Support Ubuntu-Mate/Kali, single touch;
- Support RetroPie
- Equipped with backlight brightness adjustment button

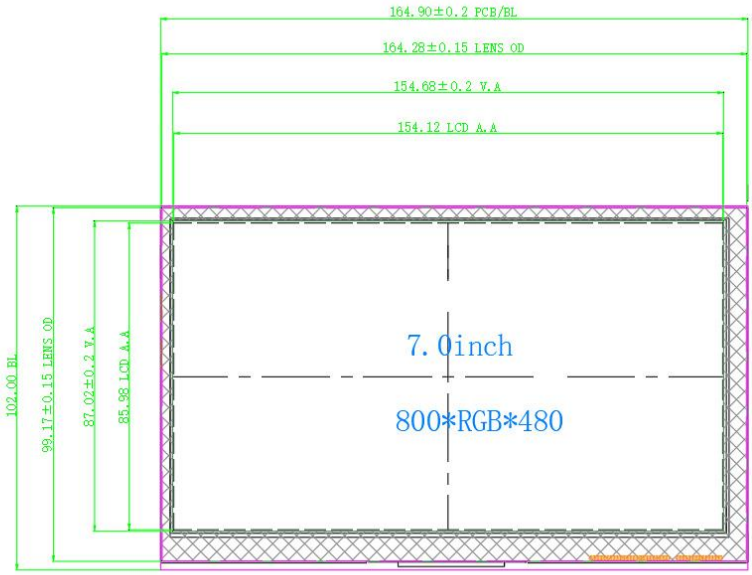
2. Parameter

Name	Explain
LCD Size	7inch
LCD Resolution	800x480
Screen Material	IPS
LCD Interface	RGB888
Module Interface	Raspberry Pi MIPI DSI
Refresh rate	60Hz
Touch screen type	Capacitive
Viewing Angle	Full perspective
Working Voltage	5V
Maximum operating current	510mA
Maximum Brightness	200cd/㎡
Working Temperature	-20℃-70℃
Module Size	164.89*102.00*12.25(mm)
Active Area	154.68*87.02(mm)

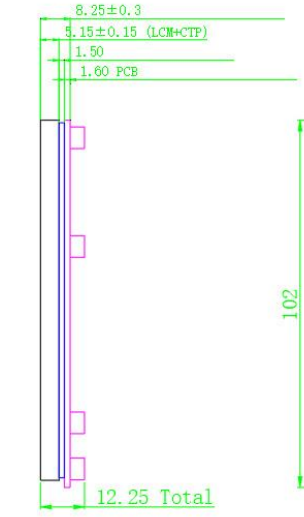
Packaging Size	212*149*54(mm)
Module Weight (net weight)	221g
Packaging Weight	330g

3. Dimensional Drawing

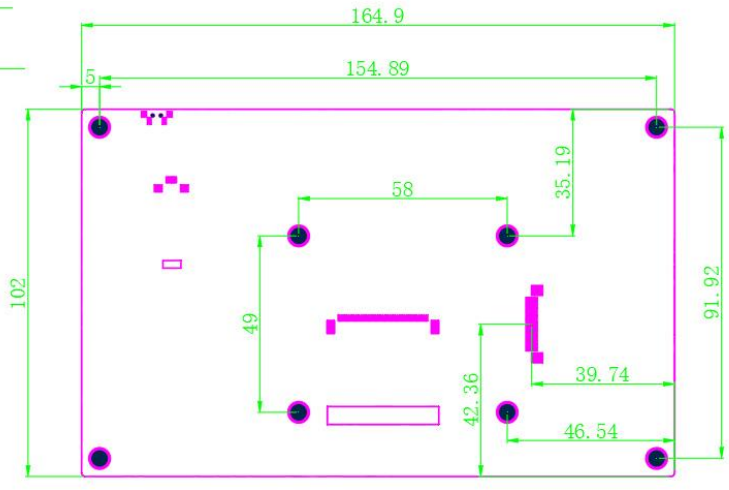
Front view



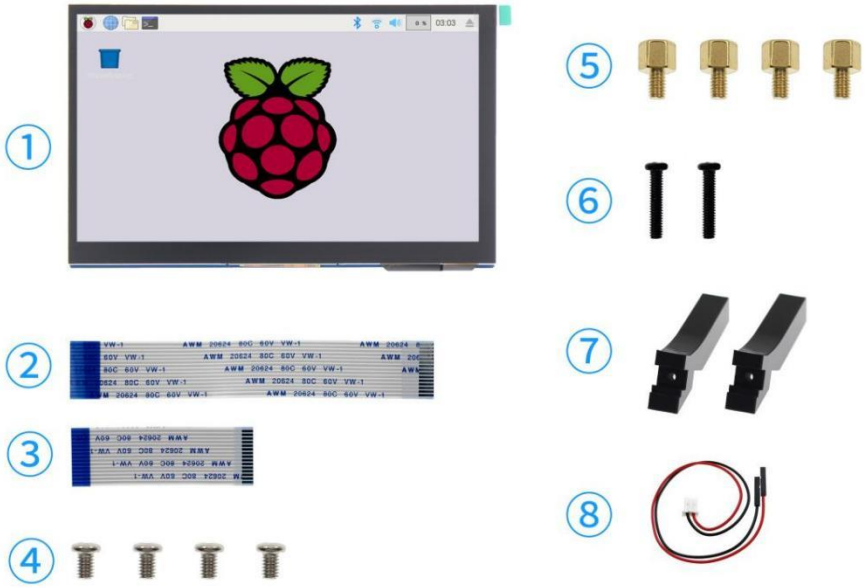
Side view



Rear view



4. Packaging List



Packaging List:

① 7 inch LCD x1	⑤ Copper Pillar (M2.5x4+4) x4
② FFC-1.0-15P-10cm (reverse) x1	⑥ Screw (M2.5x12) x2
③ FFC-1.0-15P-5cm (reverse) x1	⑦ Support x2
④ Screw (M2.5x4) x4	⑧ 2P Wire Rod x2

5. Hardware Description

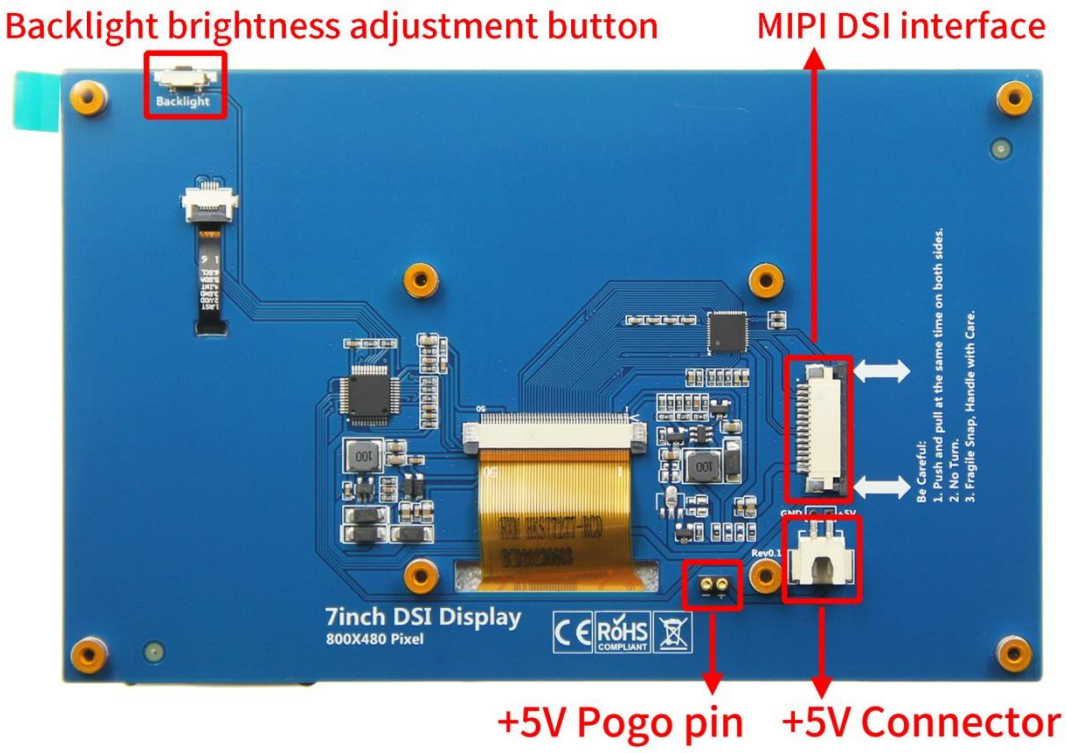


Figure 1. Hardware Interface

Detailed explanation:

(1) **Backlight brightness adjustment button:** Short press backlight change 10%, long press 3 seconds to close backlight; short press to open backlight.

(2) **MIPI DSI interface：** The specification is 15PIN with a spacing of 1.0MM; MIPI DSI interface for connecting Raspberry Pi through FPC flexible ribbon cable

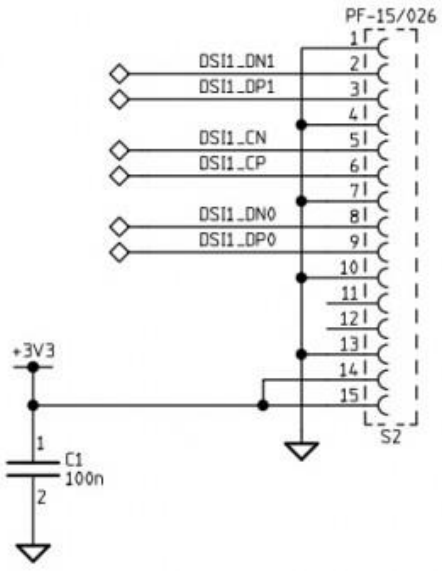


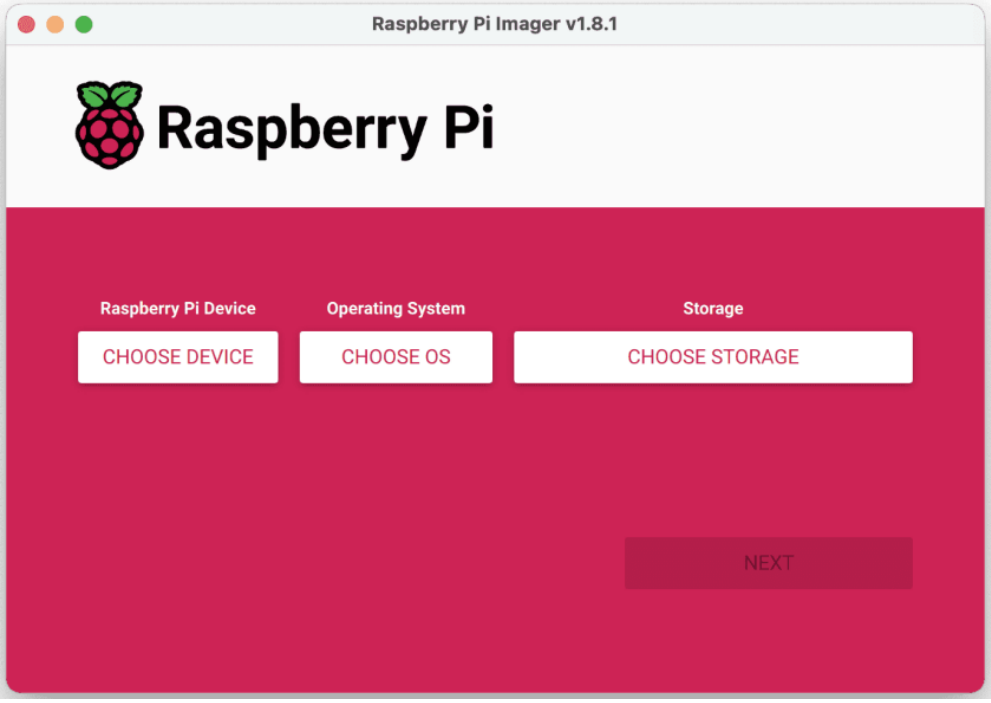
Figure 2. MIPI DSI 15PIN Pin Definition

(3) **+5V Pogo pin：** Used to power from below the Raspberry Pi pin (Raspberry Pi needs to be fixed to the module with screws and copper pillars)

(4) **+5V Connector：** Used to power from above the raspberry pie pin through the DuPont cable

6. How to use it

（1）Install Raspberry Pi OS according to the official tutorial steps, and put the Micro SD card into the Raspberry Pi.



(2) Fix the Raspberry Pi to the copper pillar with 4 screws (M2.5), then connect the FFC flexible ribbon cable to the screen (as shown in Figures 3 and 4), connect it to the Raspberry Pi MIPI DSI interface (as shown in Figure 5), and finally connect the Raspberry Pi to the power cord;

After the Raspberry Pi motherboard is powered on, the display and touch functions are normal.

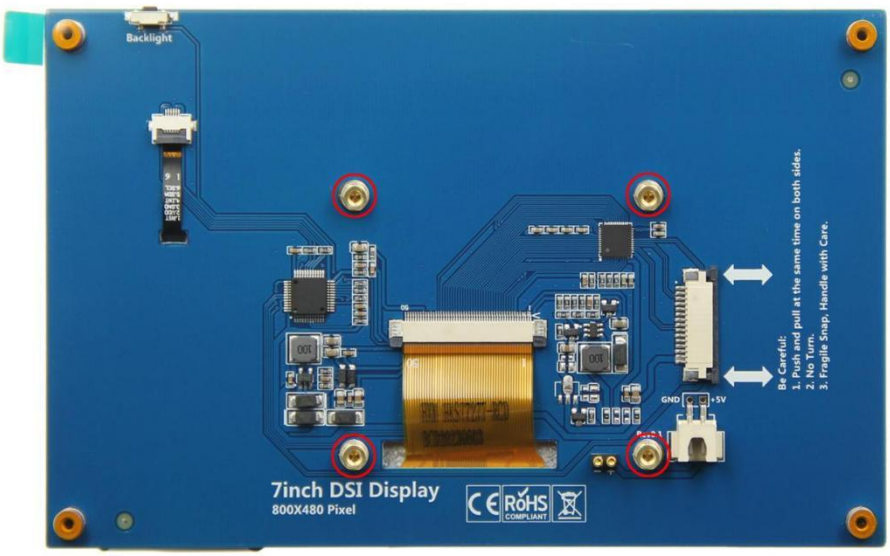


Figure 3. Installing copper pillars

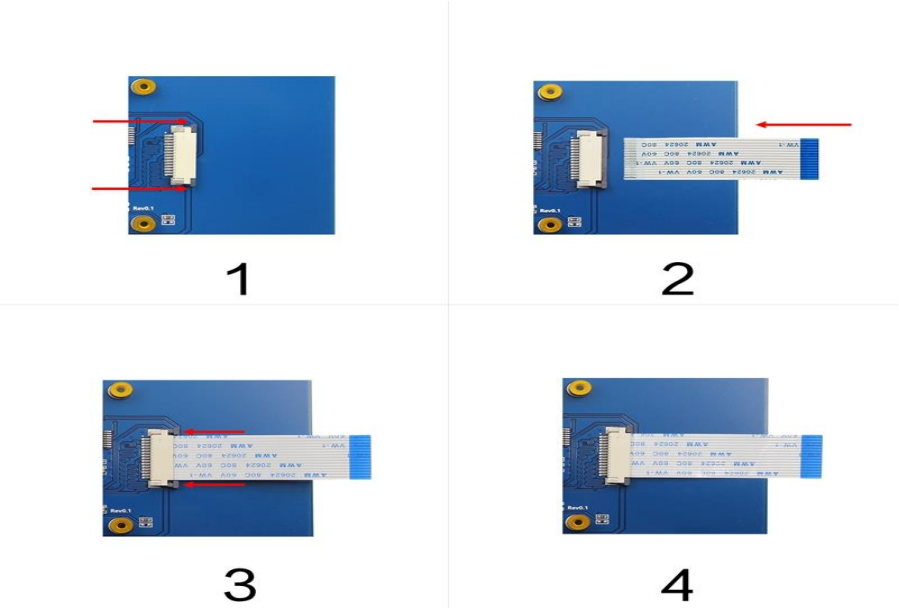


Figure 4. Connecting FFC

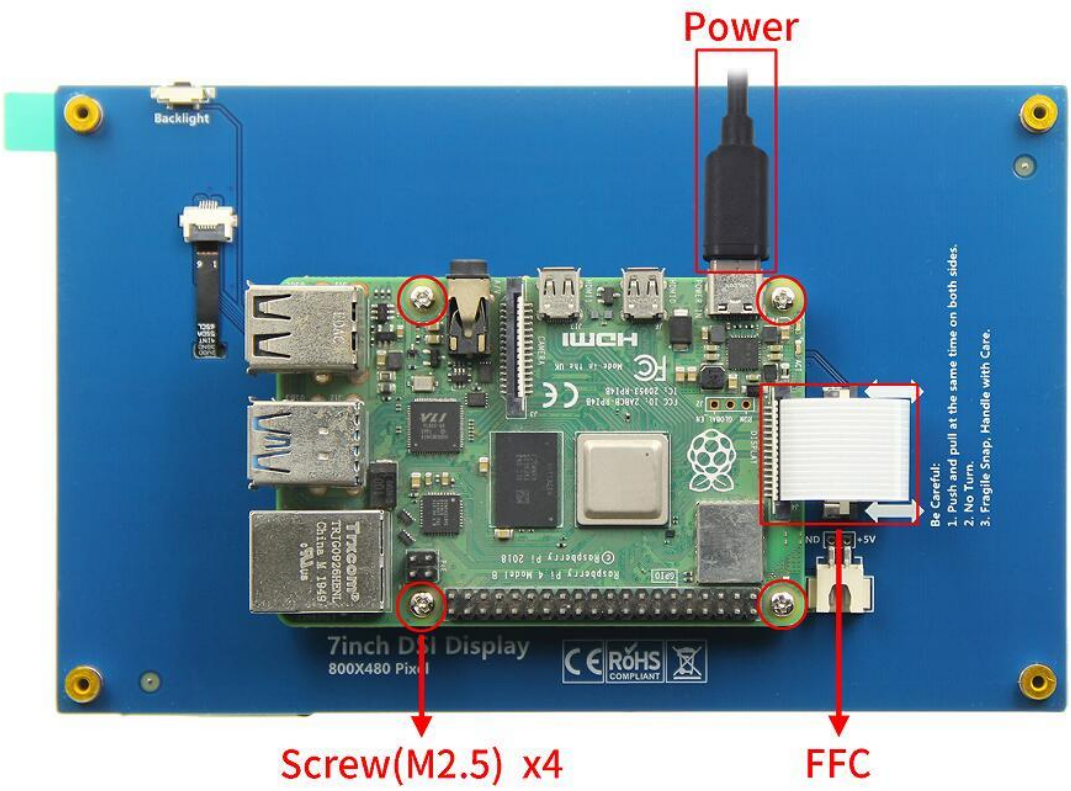


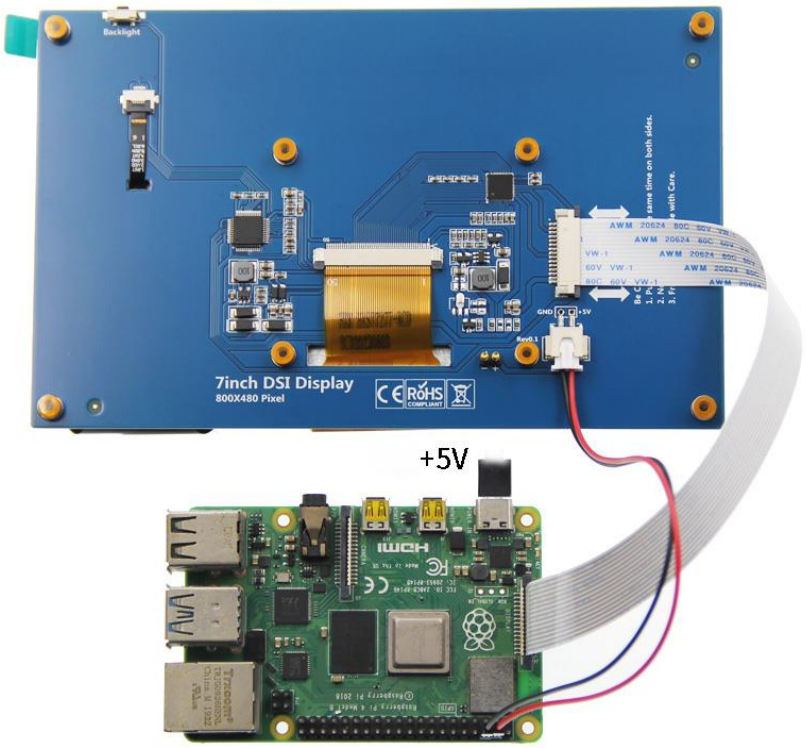
Figure 5 Connecting Raspberry Pi

Attention: The copper pillars, screws, and 15 PIN FPC cables mentioned above are all included in the product accessories.

How to connect Raspberry Pi, Method 2

Attention: When using this method of connection, a 2P custom DuPont cable must be used to connect pins 4 and 6 of the Raspberry Pi to power the module, otherwise it may damage the module!!!

If you do not want the Raspberry Pi to be fixed on the module, you can also connect it using the following method:



wiringPi 编码	BCM 编码	功能名	物理引脚 BOARD编码		功能名	BCM 编码	wiringPi 编码
		3.3V	1	2	5V		
8	2	SDA.1	3	4	5V		
9	3	SCL.1	5	6	GND		
7	4	GPIO.7	7	8	TXD	14	15
		GND	9	10	RXD	15	16
0	17	GPIO.0	11	12	GPIO.1	18	1
2	27	GPIO.2	13	14	GND		
3	22	GPIO.3	15	16	GPIO.4	23	4
		3.3V	17	18	GPIO.5	24	5
12	10	MOSI	19	20	GND		
13	9	MISO	21	22	GPIO.6	25	6
14	11	SCLK	23	24	CE0	8	10
		GND	25	26	CE1	7	11
30	0	SDA.0	27	28	SCL.0	1	31
21	5	GPIO.21	29	30	GND		
22	6	GPIO.22	31	32	GPIO.26	12	26
23	13	GPIO.23	33	34	GND		
24	19	GPIO.24	35	36	GPIO.27	16	27
25	26	GPIO.25	37	38	GPIO.28	20	28
		GND	39	40	GPIO.29	21	29

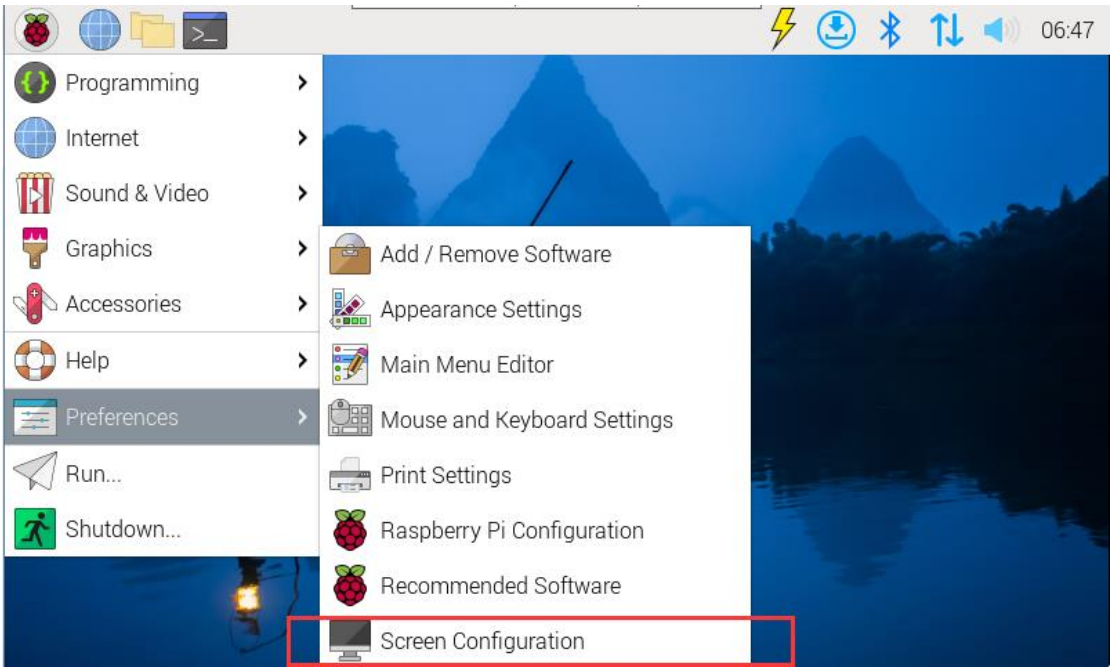
Figure 6 Connecting Raspberry Pi

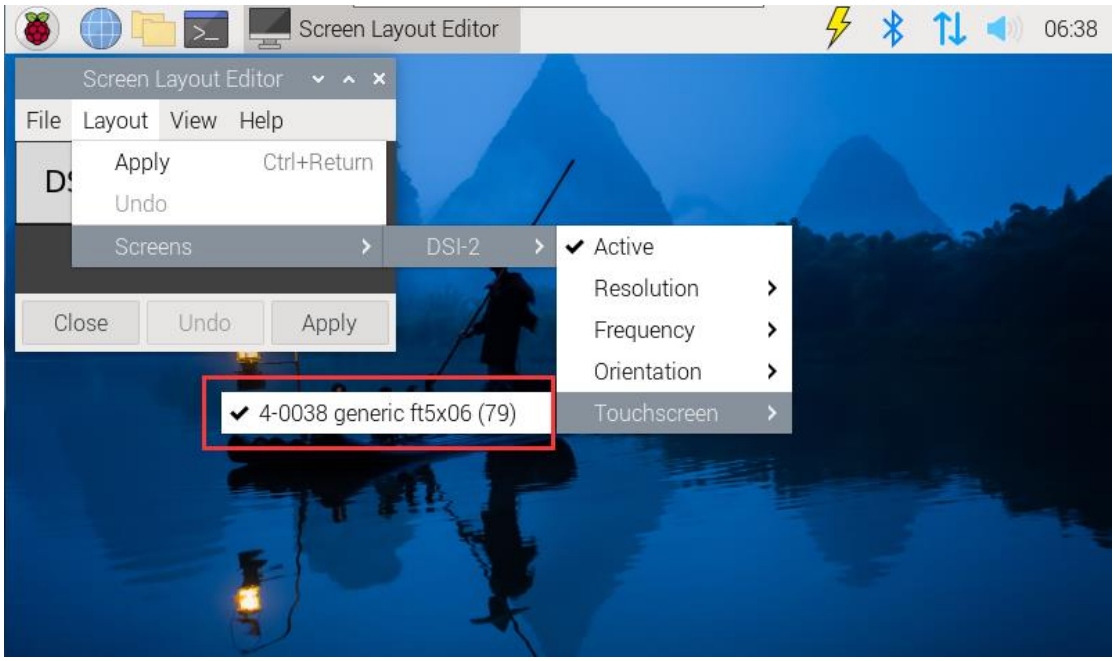
7. How to adjust brightness

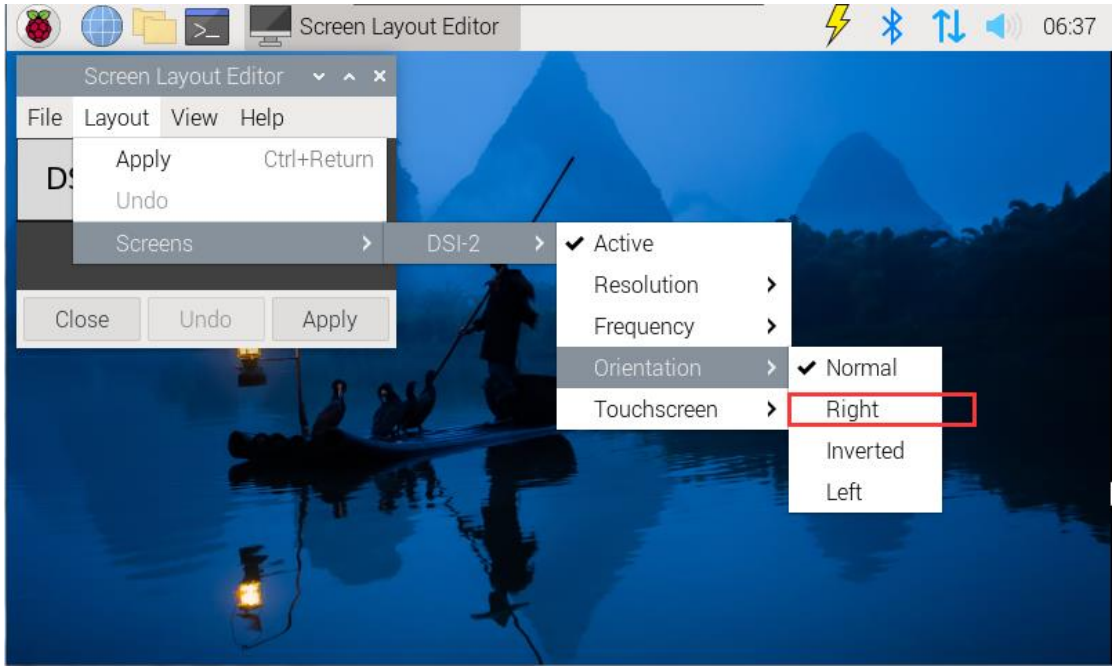
With each press of the button, the backlight brightness increases by 10%, reaching a maximum of 100%, and then cycles from 10% onwards; Long press for 3 seconds to turn off the backlight, then short press once to activate the backlight.

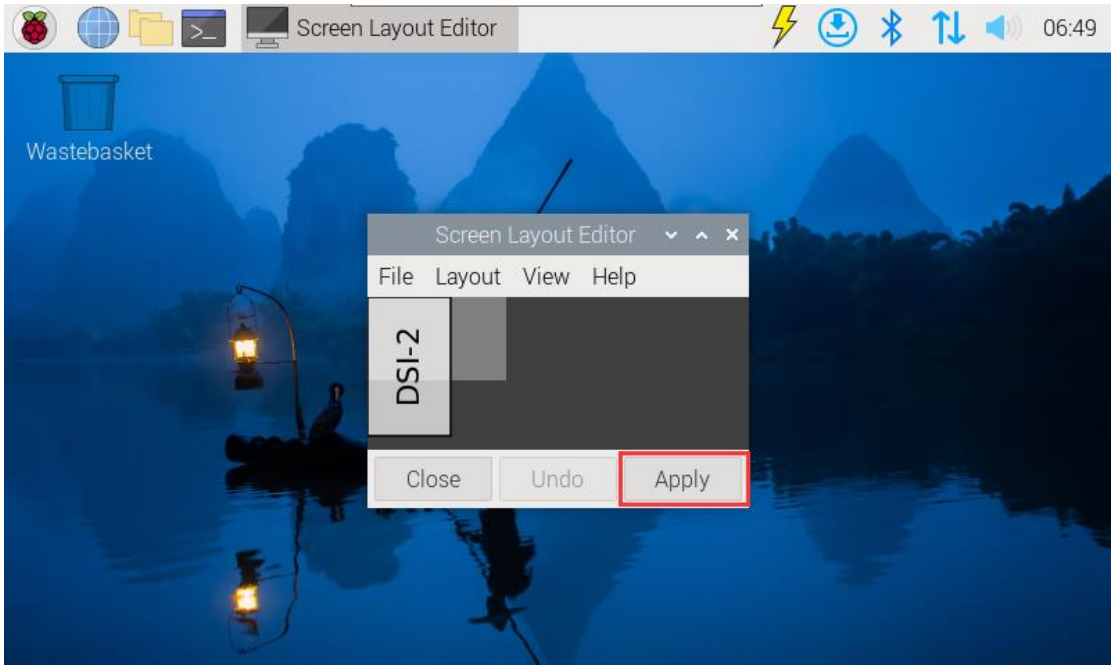
8. Display direction rotation

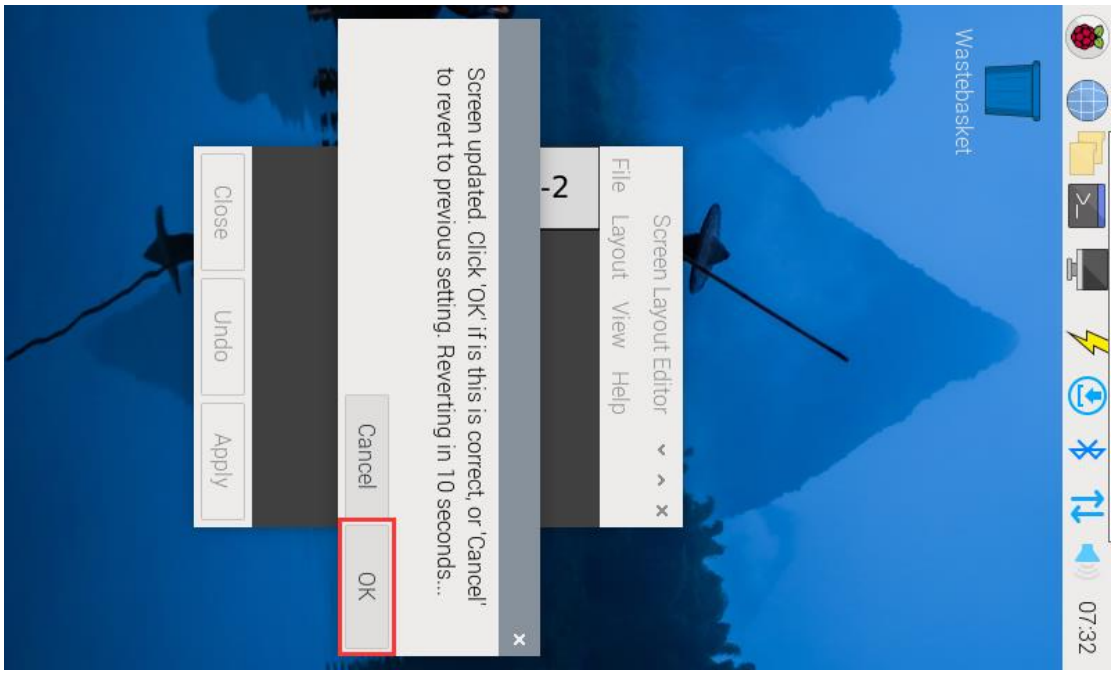
Following the method below, both the display and touch are rotated 90° clockwise. (Only for official systems after 2023-12-05, older versions of the system are not applicable)

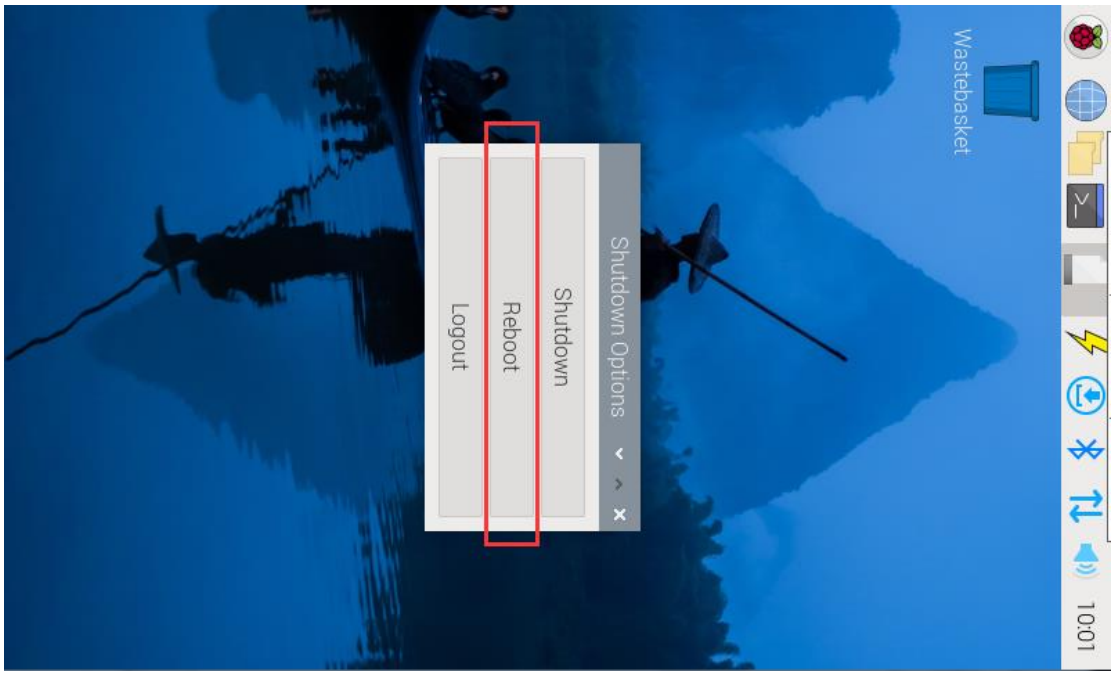



Screen Layout Editor
06:38
Screen Layout Editor
File
Layout
View
Help
Apply
Ctrl+Return
Undo
Screens
DSI-2
Active
Resolution
Frequency
Orientation
Touchscreen
Close
Undo
Apply
4-0038 generic ft5x06 (79)


Screen Layout Editor
06:37
Screen Layout Editor
File
Layout
View
Help
Apply
Ctrl+Return
Undo
Screens
DSI-2
Active
Resolution
Frequency
Orientation
Touchscreen
Close
Undo
Apply
Normal
Right
Inverted
Left


Screen Layout Editor
06:49
Wastebasket
Screen Layout Editor
File Layout View Help
DSI-2
Close
Undo
Apply


Wastebasket
Screen updated. Click 'OK' if is this is correct, or 'Cancel' to revert to previous setting. Reverting in 10 seconds...
Cancel
OK
Close
Undo
Apply
07:32



9. How to install a virtual keyboard

Execute the following command：

```
sudo apt-get install matchbox-keyboard
```